Case Study 42: Student Marks Matrix Analysis

1. Problem Statement

Process a two-dimensional grid (matrix) of academic evaluation scores across multiple students and subjects, calculating individual student total marks, individual averages, and overall subject-wise class averages.

2. Objectives

- Traverse 2D matrices across row (student) and column (subject) dimensions.
- Calculate cumulative totals and arithmetic means for individual students.
- Compute column-wise mean performance metrics for each academic subject across the cohort.

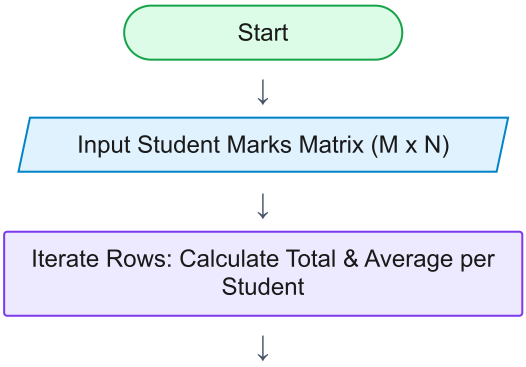
3. Assumptions

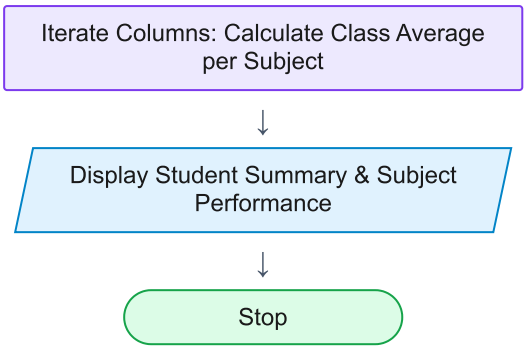
- Matrix consists of `M` rows (students) and `N` columns (subjects).
- Marks are numeric values between 0 and 100 inclusive.
- Every student has taken all scheduled subject examinations (no missing entries).

4. Algorithm

1. Start.
2. Read 2D marks matrix, student names array, and subject names array.
3. Compute row-wise totals:
   - For each student row `i`, sum elements across all subject columns `j`.
   - Calculate student average: `avg = total / N`.
4. Compute column-wise totals:
   - For each subject column `j`, sum elements across all student rows `i`.
   - Calculate subject class average: `subject_avg = column_total / M`.
5. Display formatted student scorecard table and subject average breakdown.
6. Stop.

5. Flowchart





6. Pseudocode

```
START INPUT matrix, students, subjects num_students = LENGTH(matrix) num_subjects =
LENGTH(matrix[0]) student_totals = [] student_averages = [] FOR i FROM 0 TO
num_students - 1 DO row_sum = 0 FOR j FROM 0 TO num_subjects - 1 DO row_sum =
row_sum + matrix[i][j] ENDFOR APPEND row_sum TO student_totals APPEND (row_sum /
num_subjects) TO student_averages ENDFOR subject_averages = [] FOR j FROM 0 TO
num_subjects - 1 DO col_sum = 0 FOR i FROM 0 TO num_students - 1 DO col_sum =
col_sum + matrix[i][j] ENDFOR APPEND (col_sum / num_students) TO subject_averages
ENDFOR PRINT student_totals, student_averages PRINT subject_averages STOP
```

7. Test Case and Expected Output

Evaluation Matrix Output (3 Students × 3 Subjects):

```
Student | Total | Average ------------------------------ Alice | 263 | 87.67 Bob |
240 | 80.00 Charlie | 281 | 93.67 Subject Averages: - Math: 85.00 - Physics: 89.00
- Chemistry: 87.33
```

8. Conclusion

The Student Marks Matrix Analysis tool systematically processes academic score arrays, delivering dual-dimensional analysis for both individual student evaluation and cohort subject-level insights.